



DATA VISUALIZATION DESIGN PROJECT

By Team Vizionaries

Beyond the Credit Score

Uncovering Hidden Signals of Loan Repayment Risk



Welcome to



APPROVE & PRAY

Financial Services

(Yes... that's the old way.)

For too long, loan approvals have relied on gut instinct, incomplete signals, and a bit of hope. At **Approve & Pray Financials**, that approach has cost us revenue, increased defaults, and left good customers behind.

Today, we're changing that...

Instead of approving and praying, we're building a data backed lending strategy, one that uses customer income profiles, spending behavior, credit history, and past loan performance to make smarter, fairer, and more confident decisions.



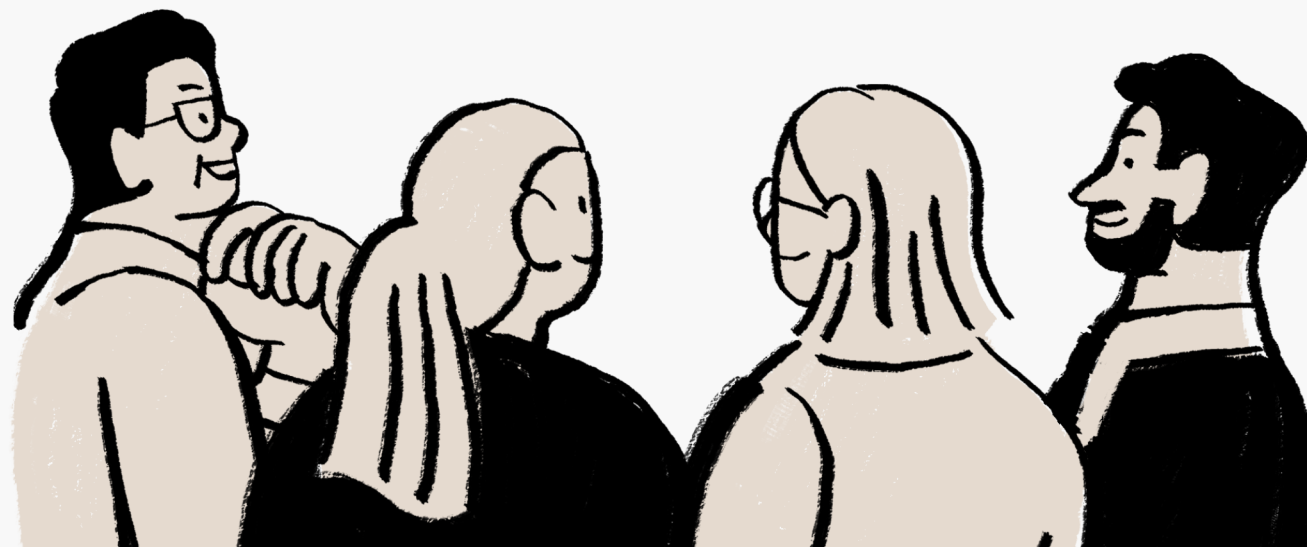
It all starts with Data

Application Data

Customer-level data captured at loan application time, including demographics, income, assets, and credit signals.

Previous Applications

Historical records of past loan applications, including approval decisions and loan terms.



Defaulter Distribution

Defaulters: 24,825 customers (8.07%)

Non-Defaulters: 282,686 customers (91.93%)

The dataset is highly imbalanced, with defaults representing a small but critical minority of cases.

Missing Information

application_data - 49 Columns Contain > 40% missing Values

previous_application - 3 Columns contain > 85% missing values

Outliers

Presence of extreme values in loan amount, and credit-related variables.

On customer - level data: Number of family members, car age, income etc.

Additional Preprocessing

XNA, XAP considered as null values.

Let's Review Where We Stand

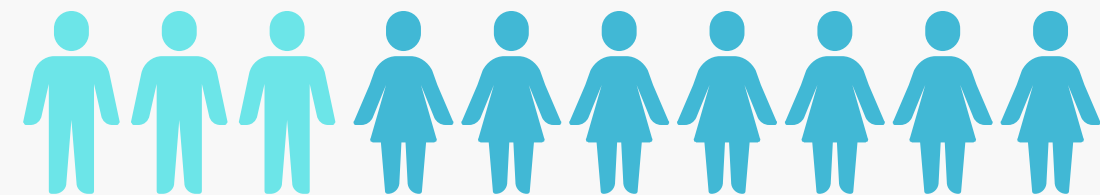
Our Customer Base:

Defaulters: ~8% (~2 in every 25)



Customer Demographics:

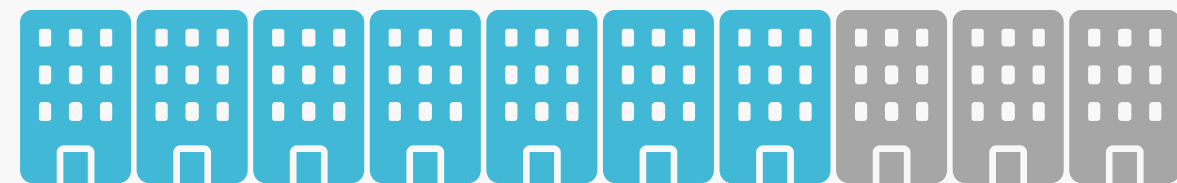
Gender: ~66% Women and ~34% Men (~7 in every 10 is a female customer)



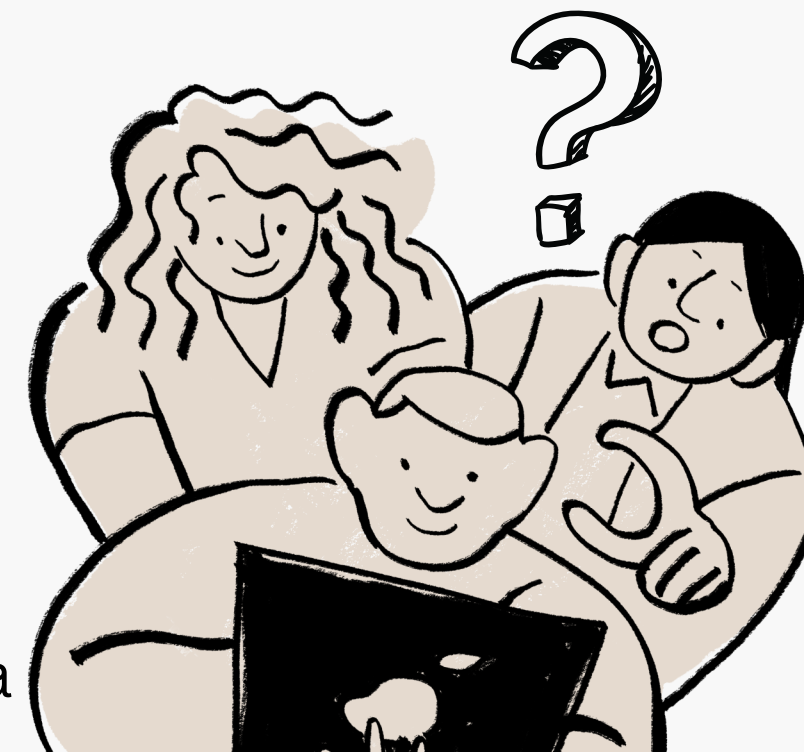
Car Ownership: 3 out of every 10 applicant owns a car.



House / Flat Ownership: 7 out of every 10 applicant owns a house or flat.



Visualization Tool Used: Canva



Let's Review Where We Stand

Loan Application Types

~90% Cash loans, ~10% on revolving loans



Who defaults?

Gender: ~10% of Males Default while ~7% of Females Default.

Car Ownership:

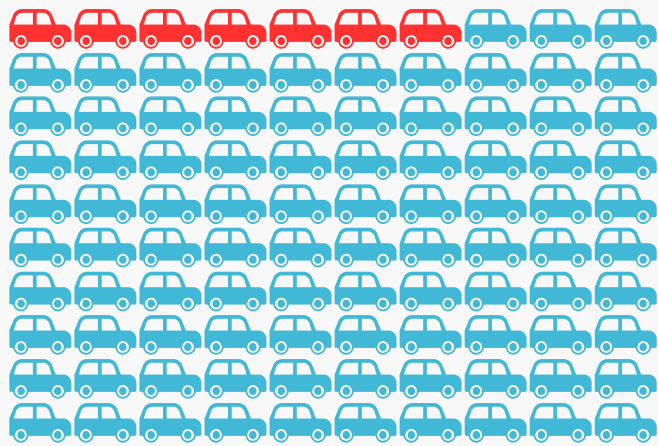
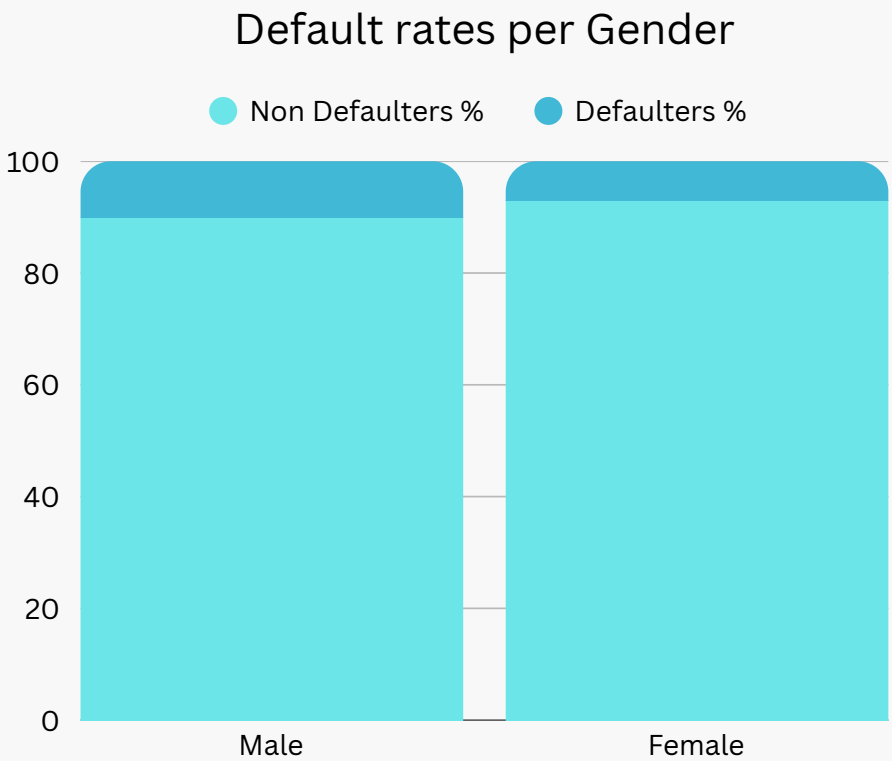
7 out of 100 car owners default.

8 out of 100 non car owners default.

House / Flat Ownership:

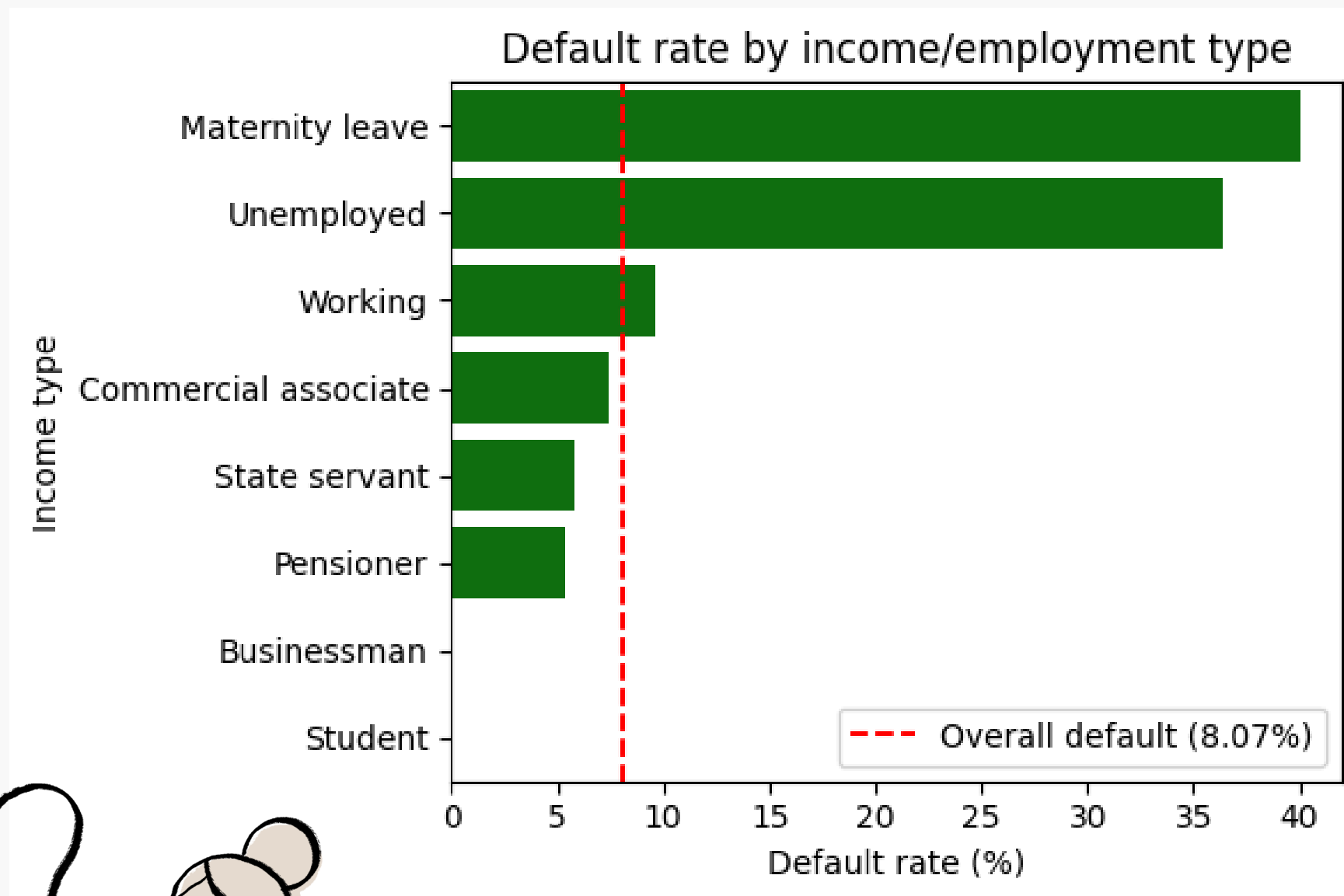
7 out of 100 house owners default.

8 out of 100 non house owners default.

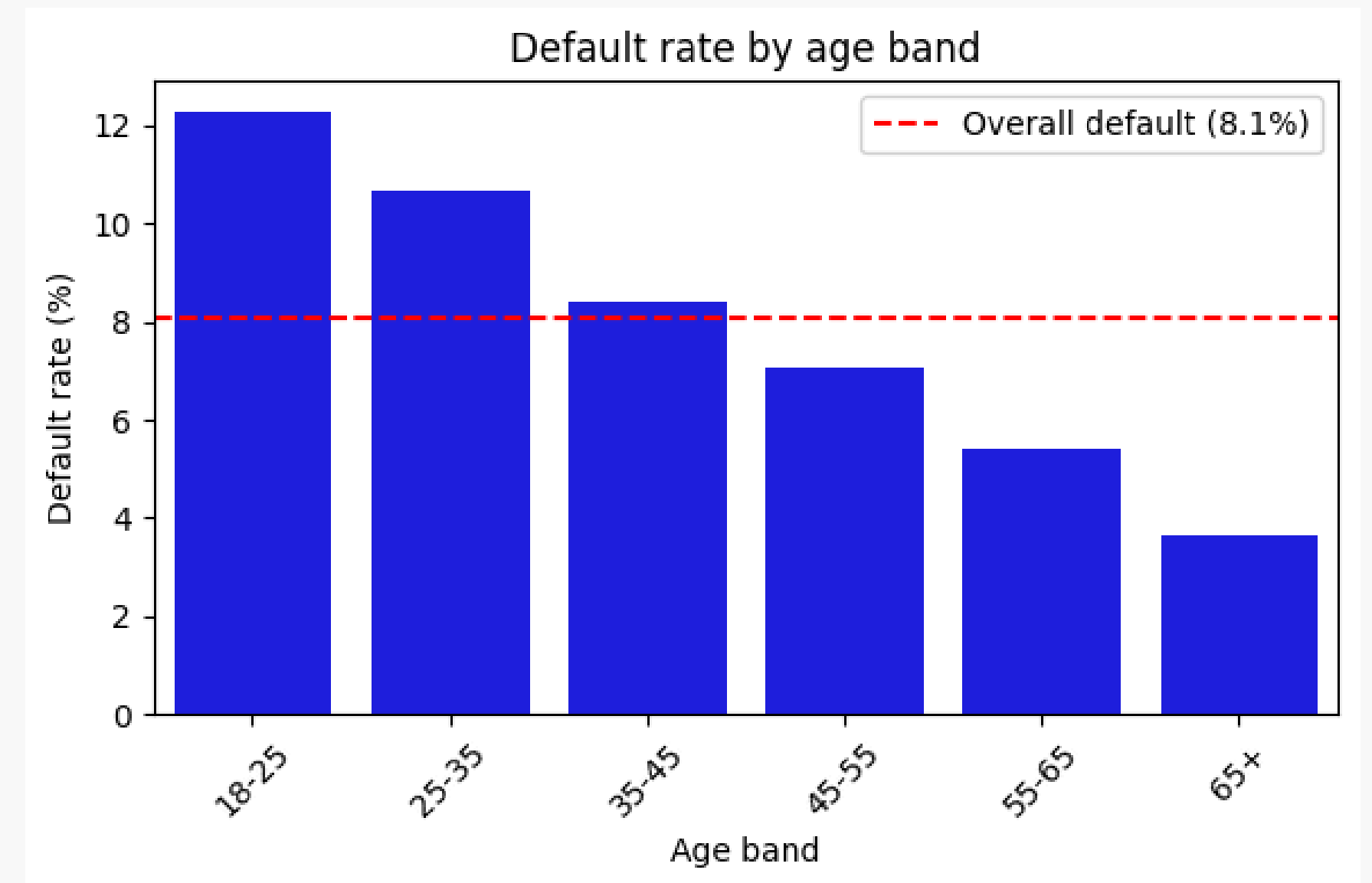


Trends & Takeaways

Can income type determine defaulters?



Can age group determine defaulters?

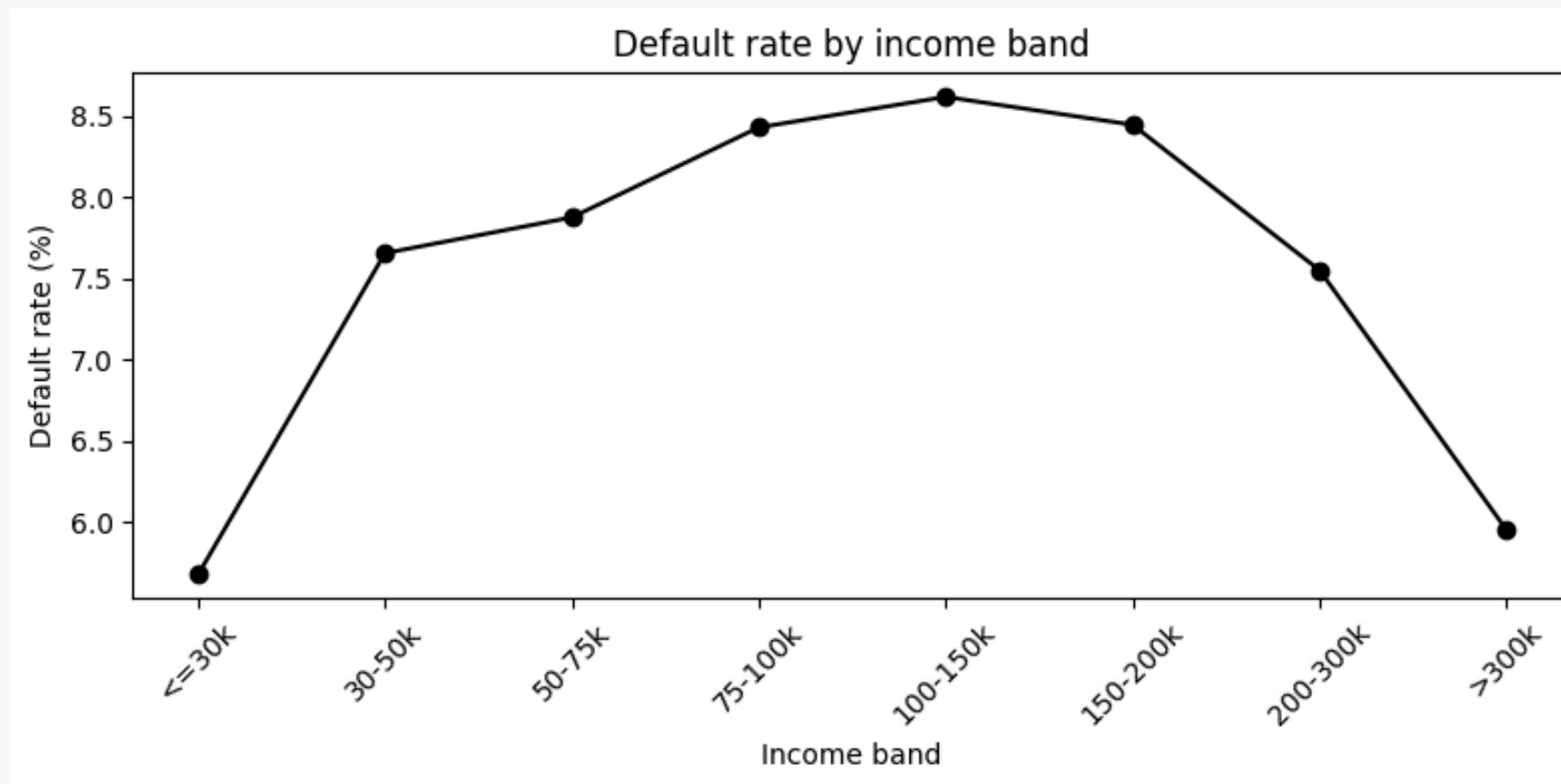


Visualization Tool Used: Matplotlib

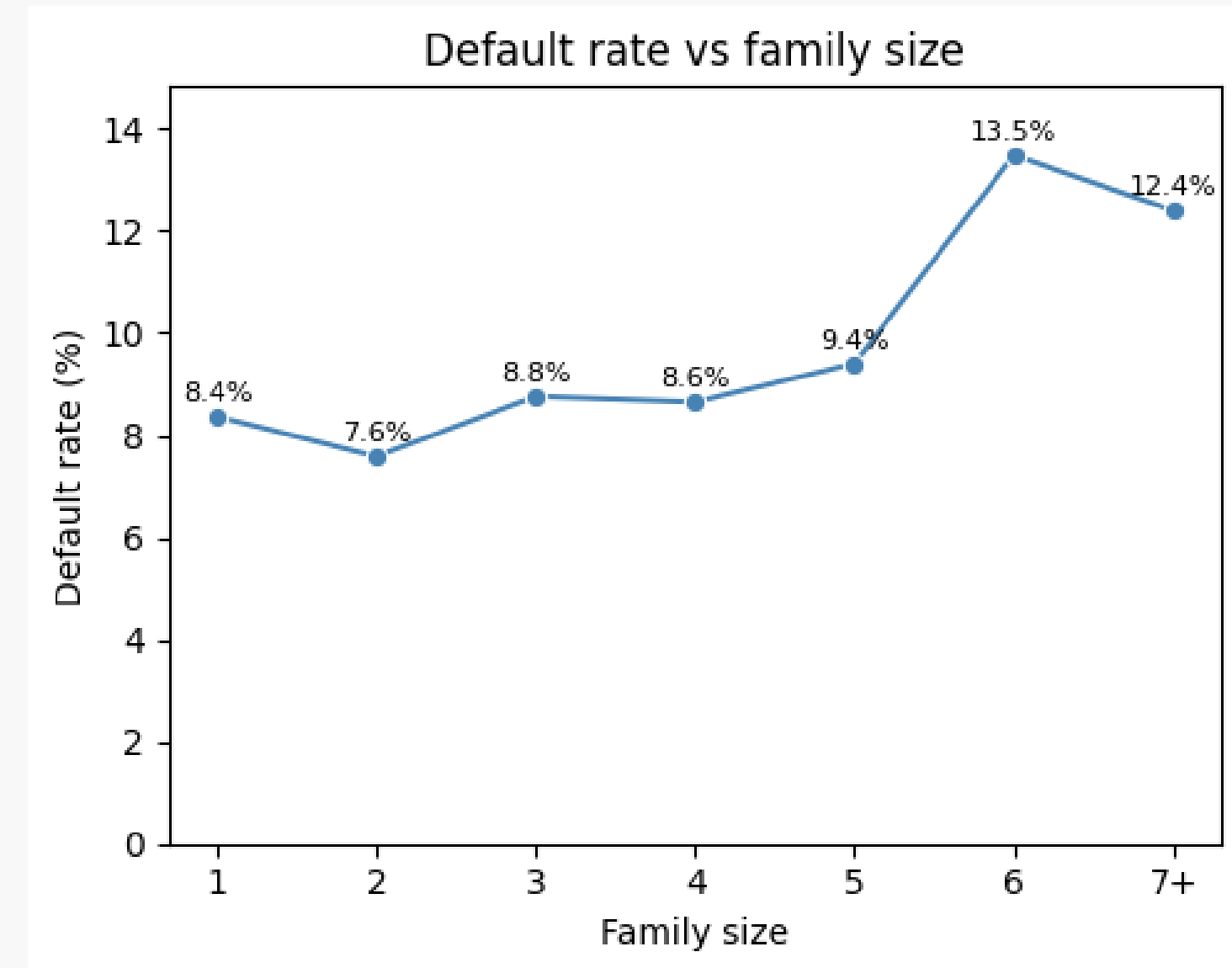


Trends & Takeaways

Can income determine defaulters?



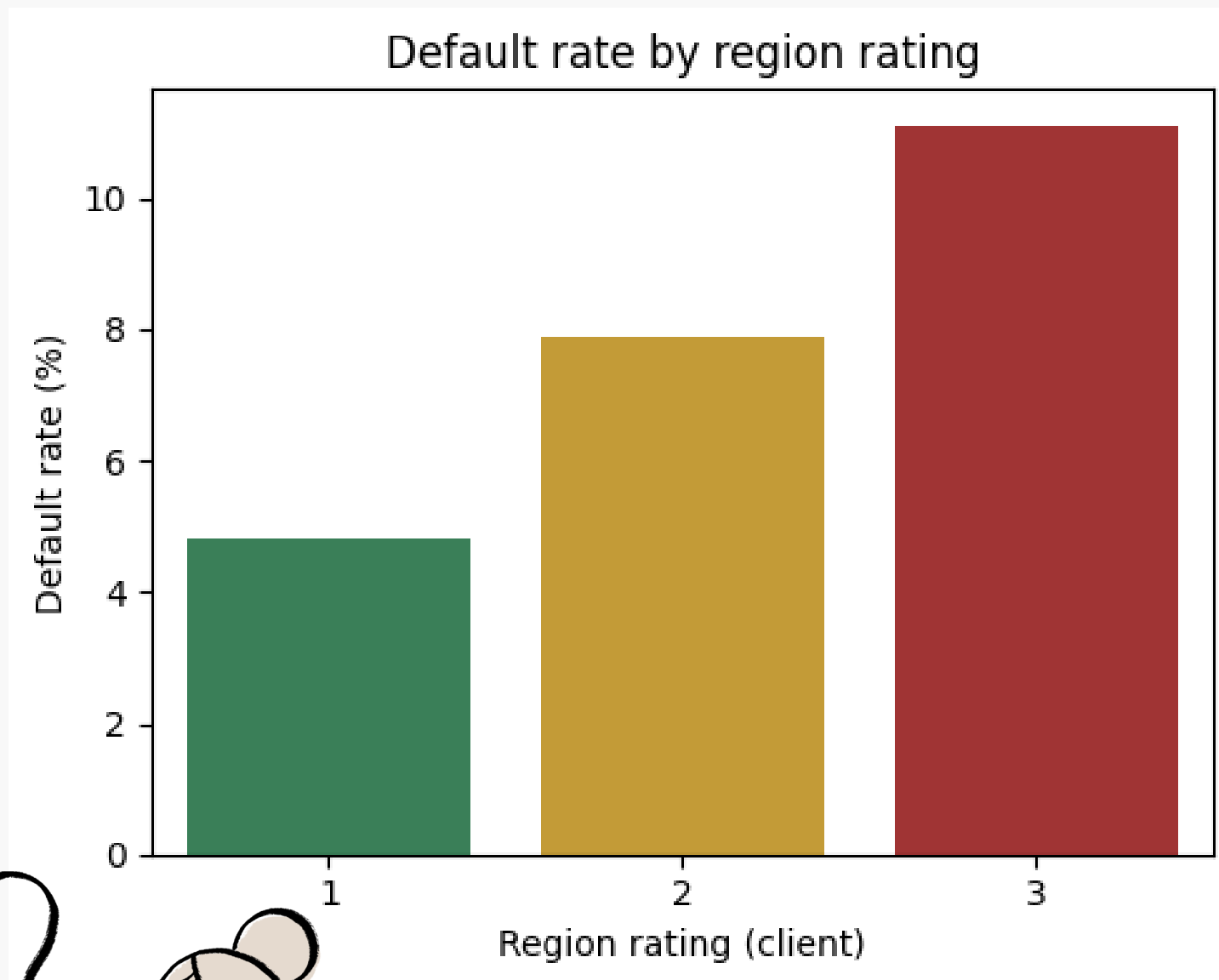
Can family size determine defaulters?



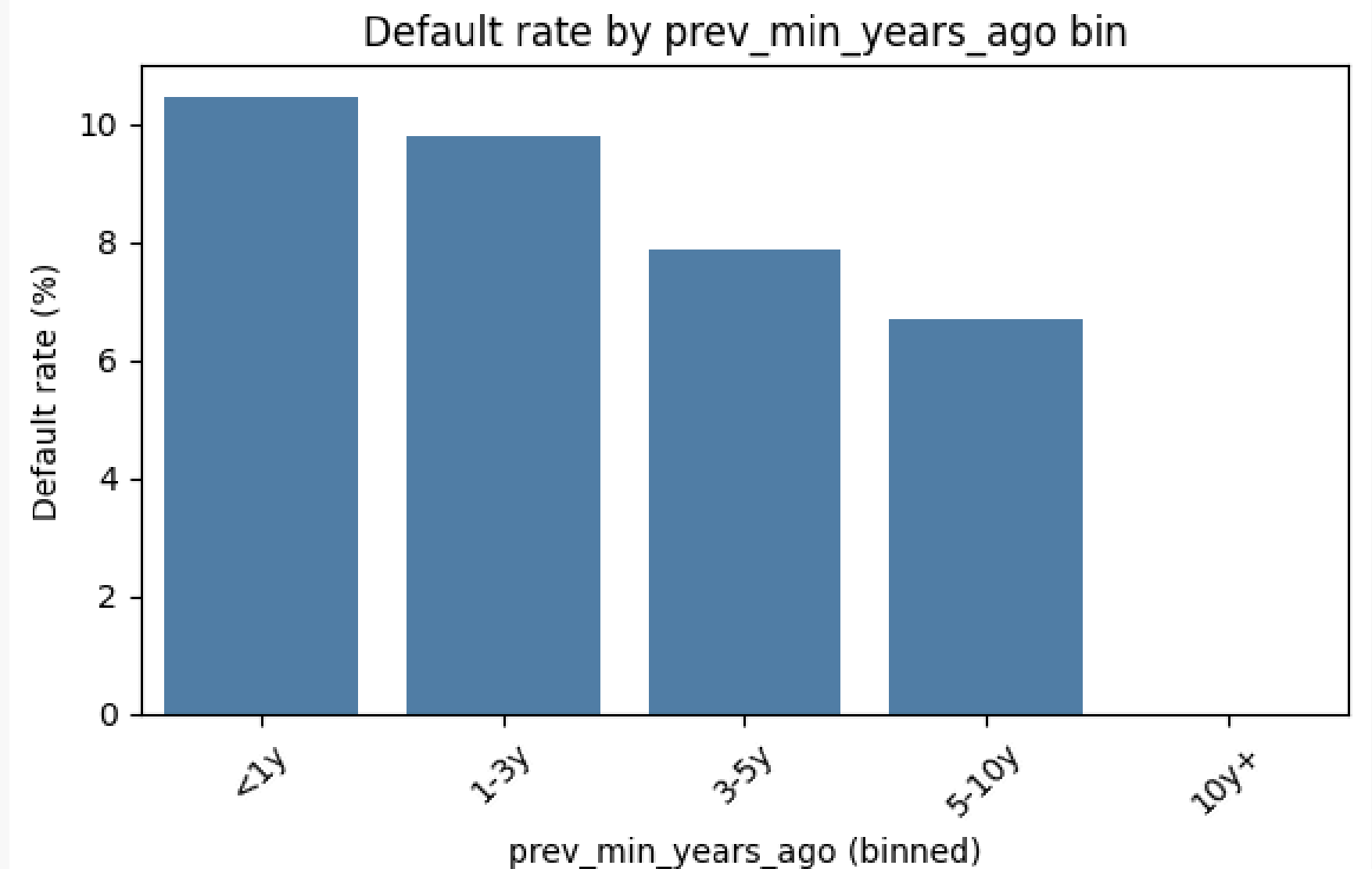
Visualization Tool Used: Matplotlib

Trends & Takeaways

Region Rating



Duration between previous min days decision and current loan application

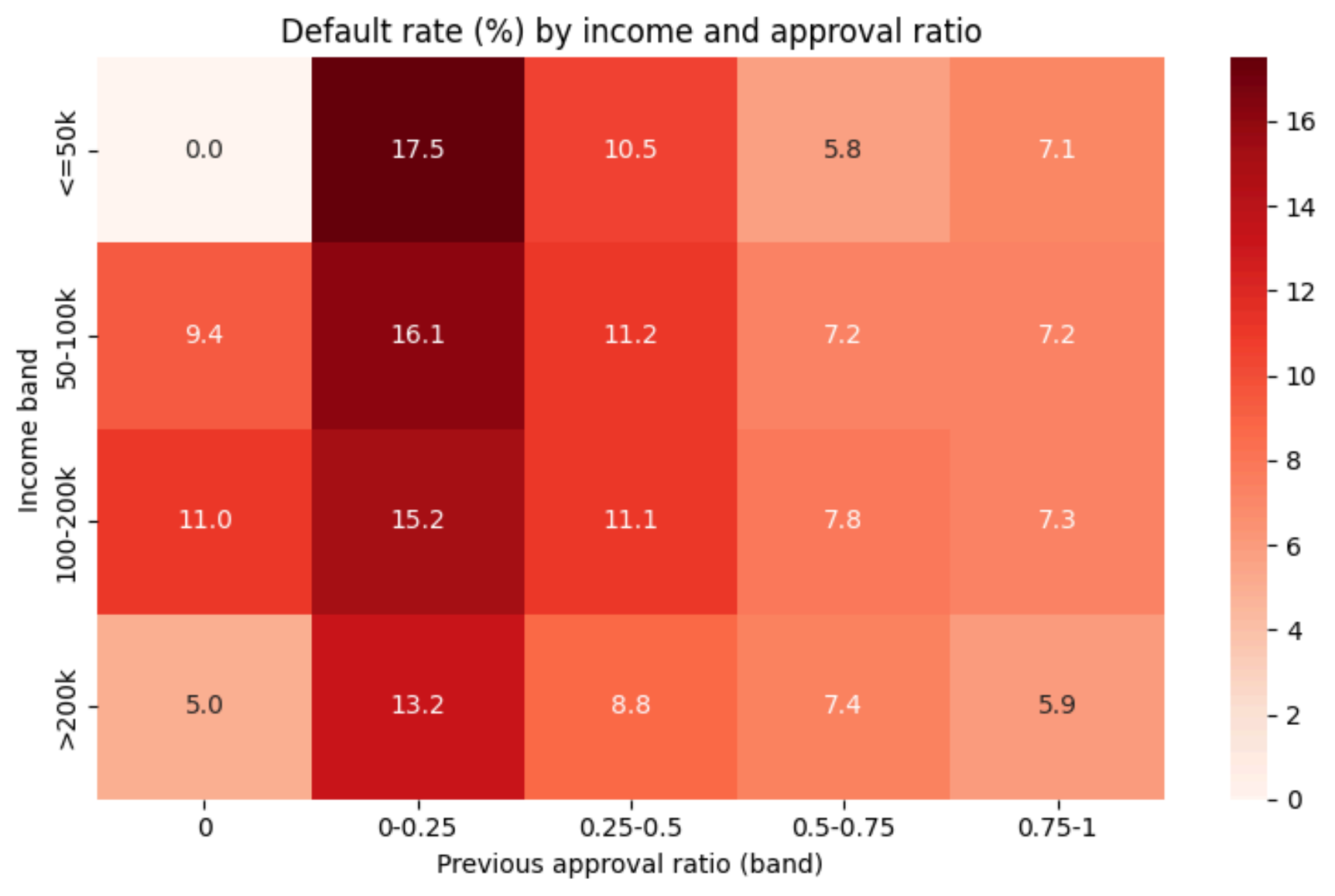


Visualization Tool Used: Matplotlib



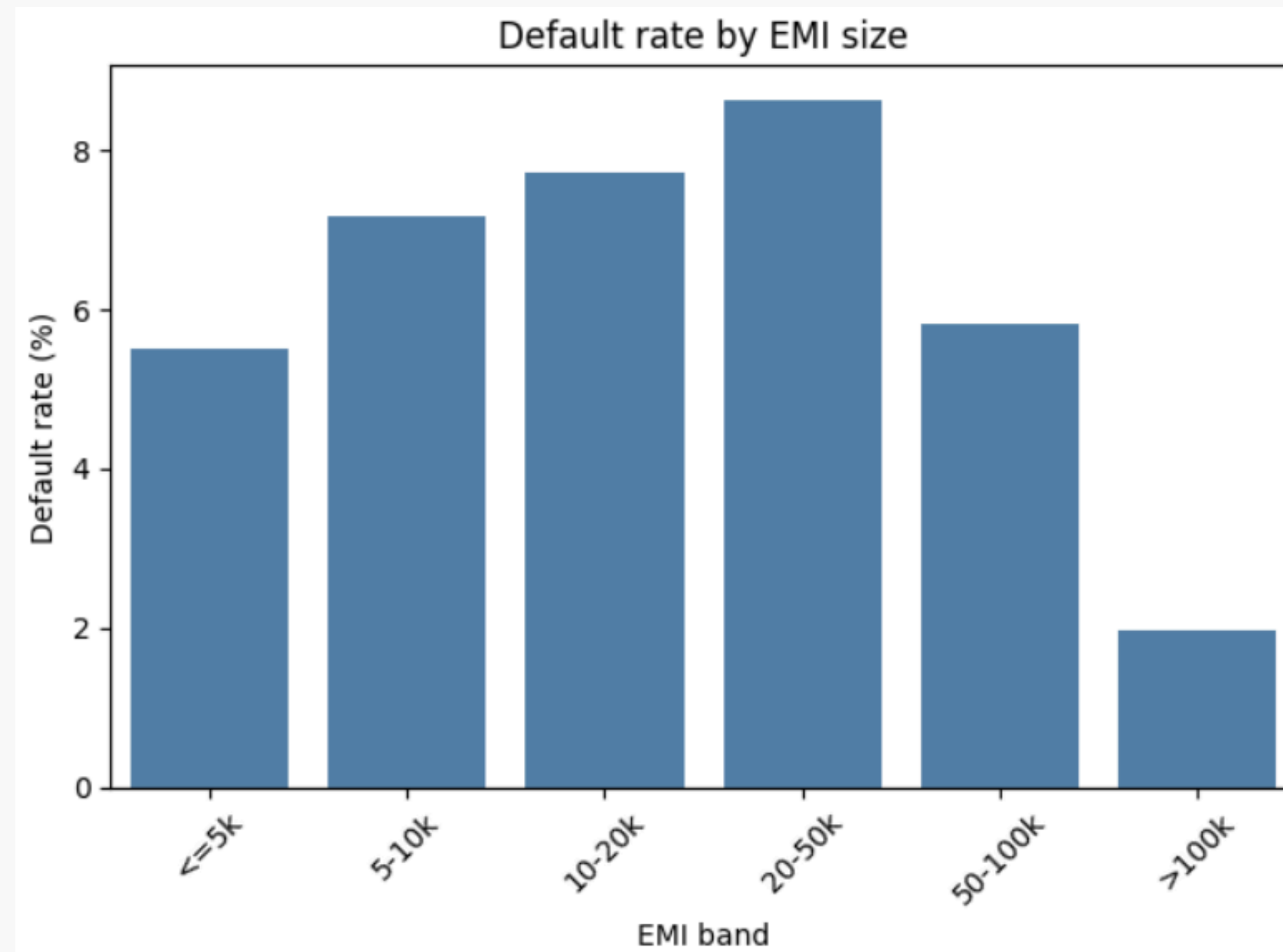
Trends & Takeaways

Income Band × Approval Ratio



Trends & Takeaways

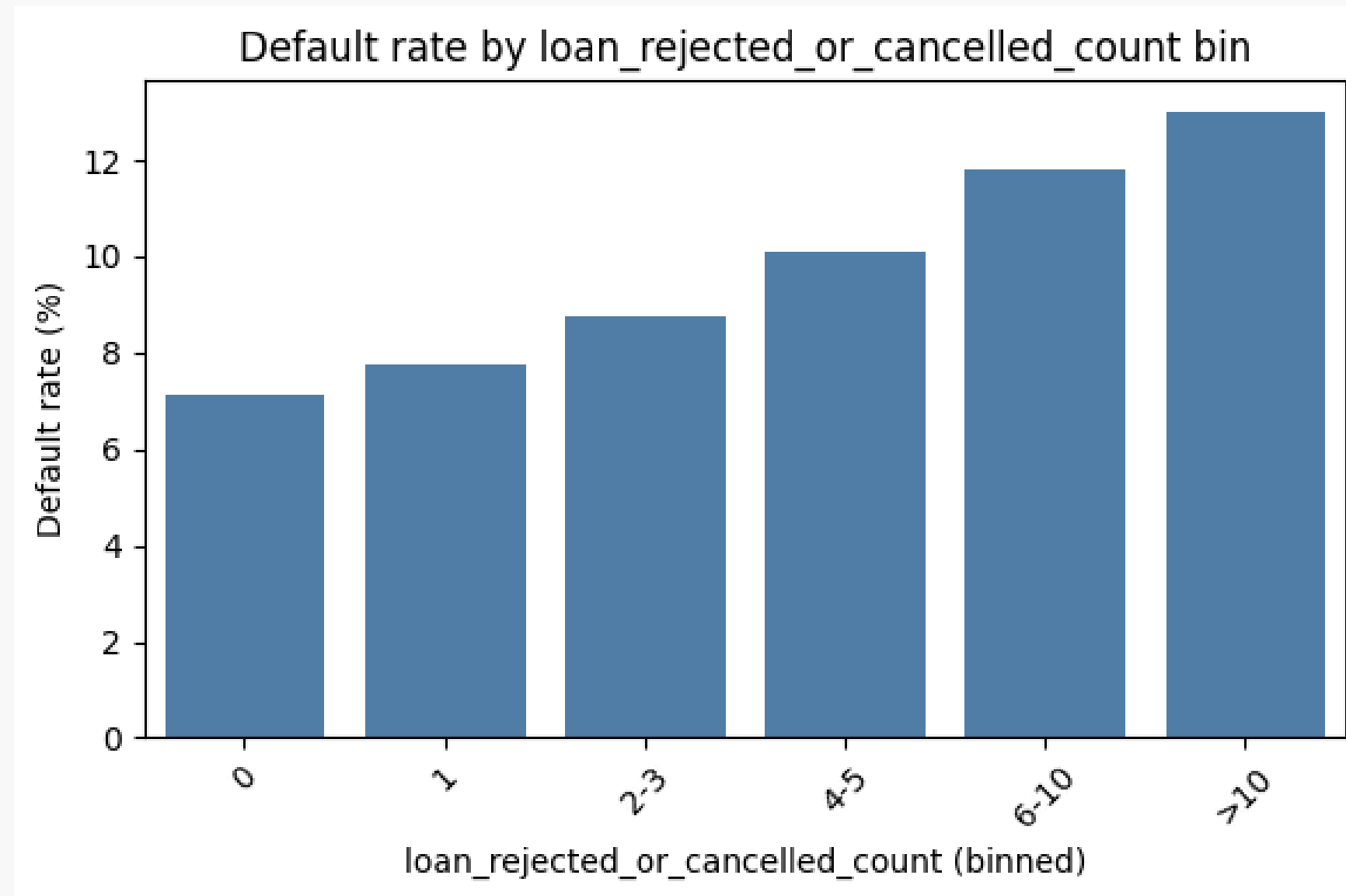
How does EMI affect Default Rates?



Visualization Tool Used: Matplotlib

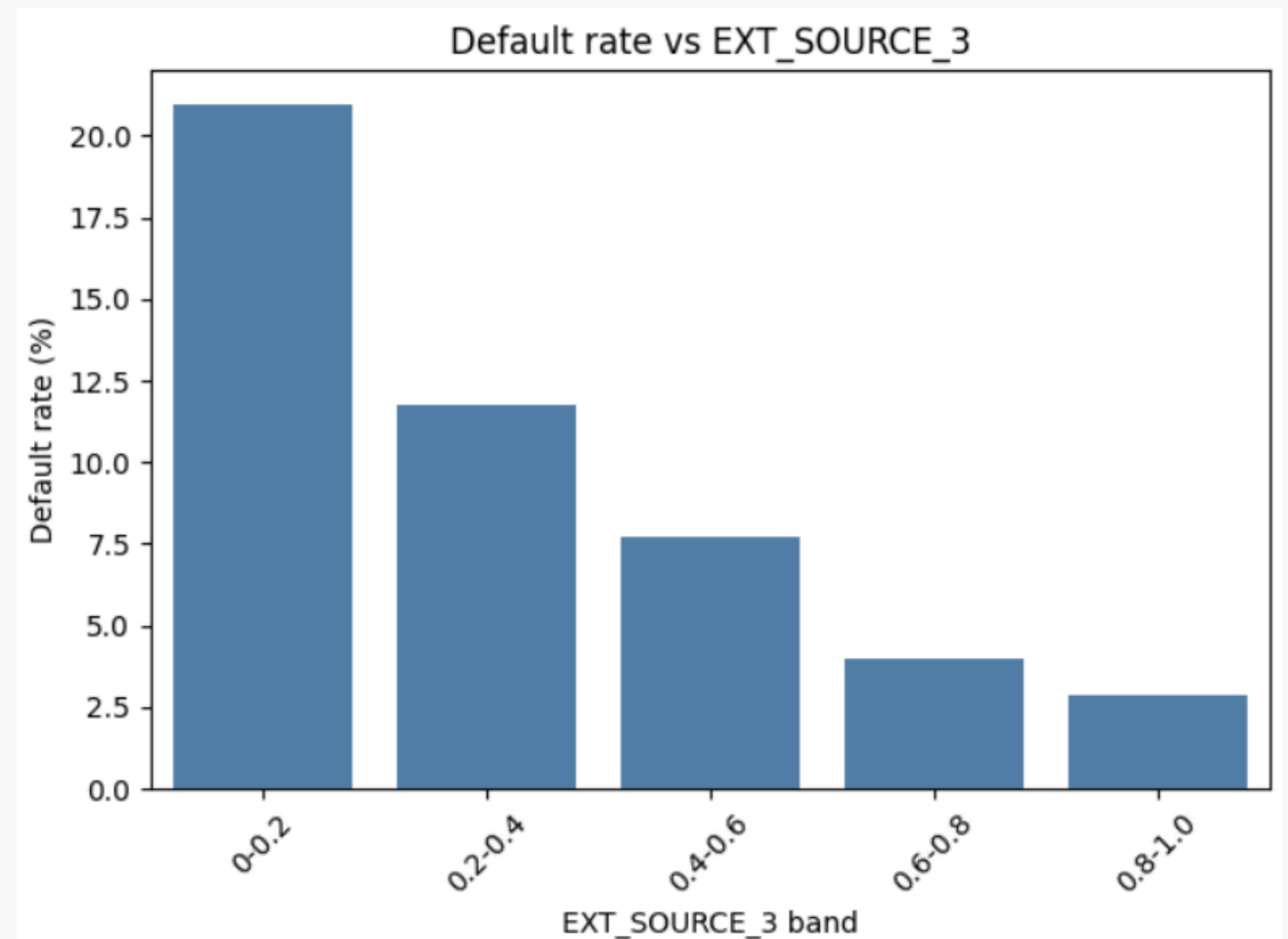
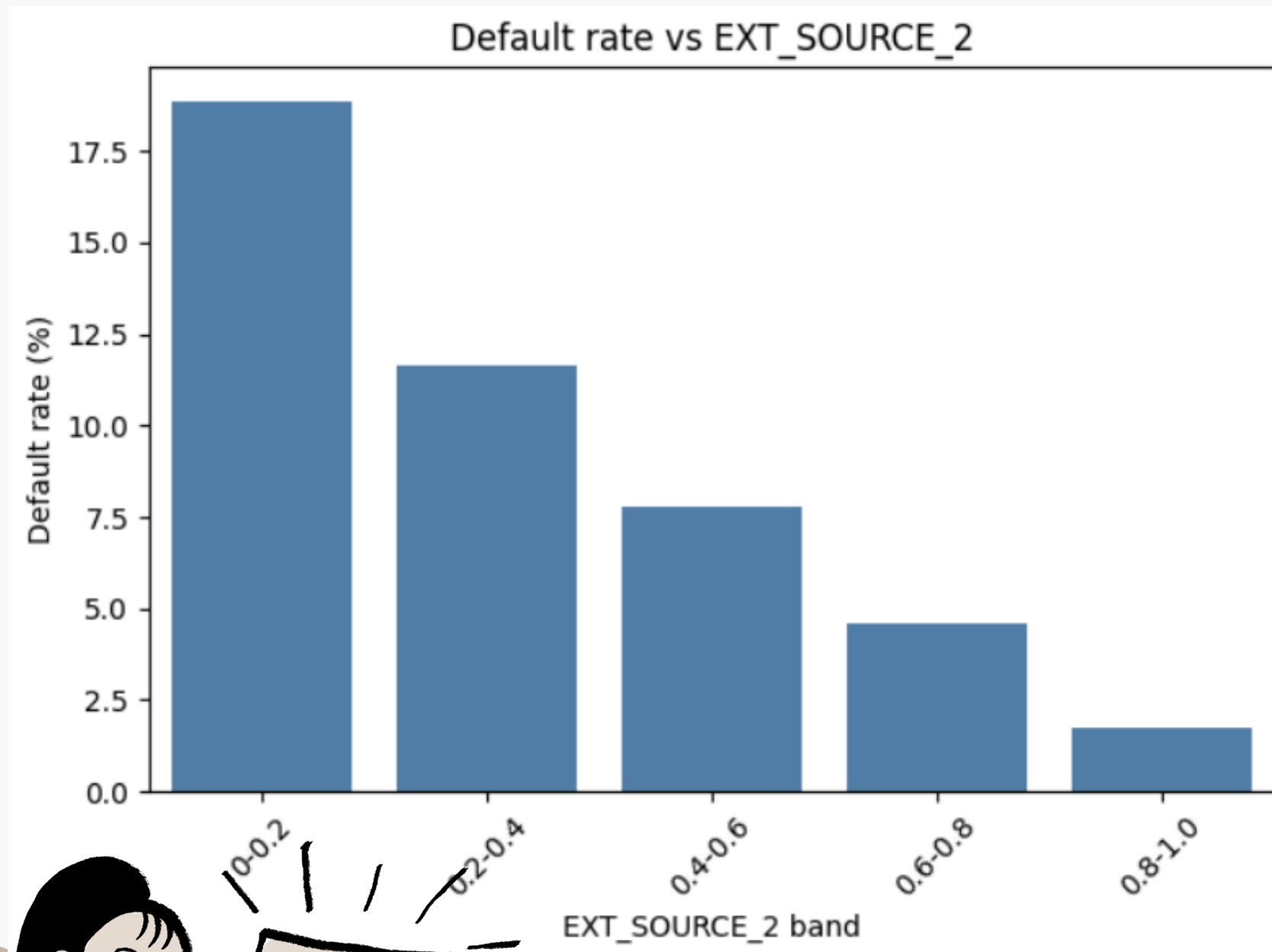
Past Behaviour

Previous loan rejections / Cancellations vs Default Rate



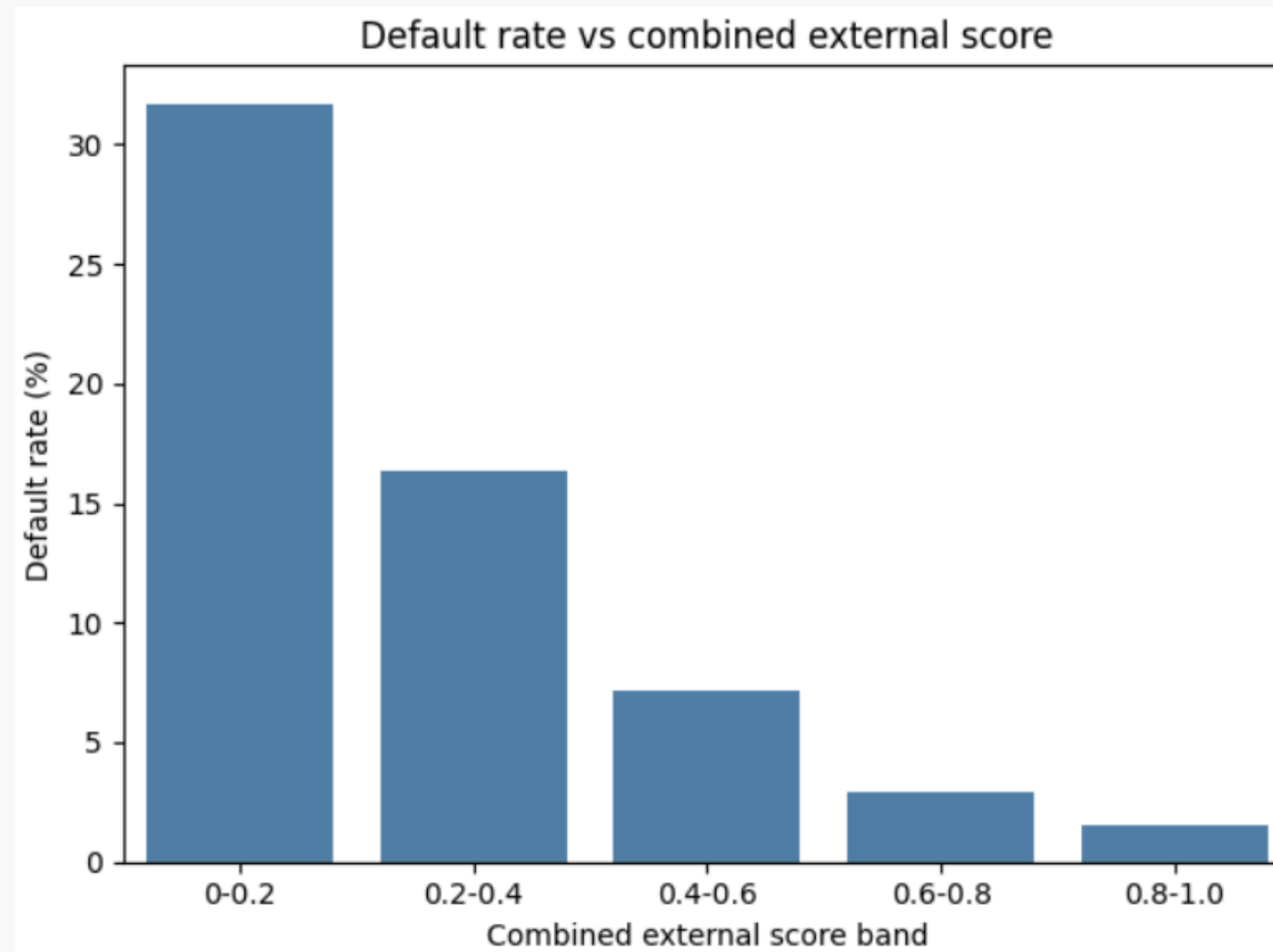
Visualization Tool Used: Matplotlib

What External Sources tell us



Visualization Tool Used: Matplotlib

A Unified View of the External Sources



Findings and insights from the data

- Unemployed applicants have previously defaulted 36% of times. This is almost 4 times higher than working applicants who default 9.6%.
- Applicants of age group between 18 - 25 have the highest default rate. The trend of age group vs defaulter rate suggests risk of defaulting reduces with increasing age.
- Applicants with Income <50k who have been approved 25% of time previously default ~17% of the time.
- Applicant with 6+ previous rejections default more than 10% of time. This is a strong behavioral signal of such applicants.
- Applicants with external score <0.3 are likely to default by 15-20% on average.
- High EMI (>50k): High risk



Plan forward (Recommendations)

- Apply stricter scrutiny for unemployed applicants.
- Consider requiring additional collateral, guarantors, or proof of alternative income before approval.
- Offer smaller loan amounts or shorter tenures to reduce exposure.
- For younger applicants, conduct additional checks on income stability and repayment capacity.
- Apply stricter affordability assessments for low-income applicants.
- Consider lower loan amounts or staggered disbursements to reduce exposure.
- Treat multiple previous rejections as a strong behavioral red flag.
- Cap EMIs relative to applicant's income (e.g., EMI should not exceed 40–50% of monthly income).

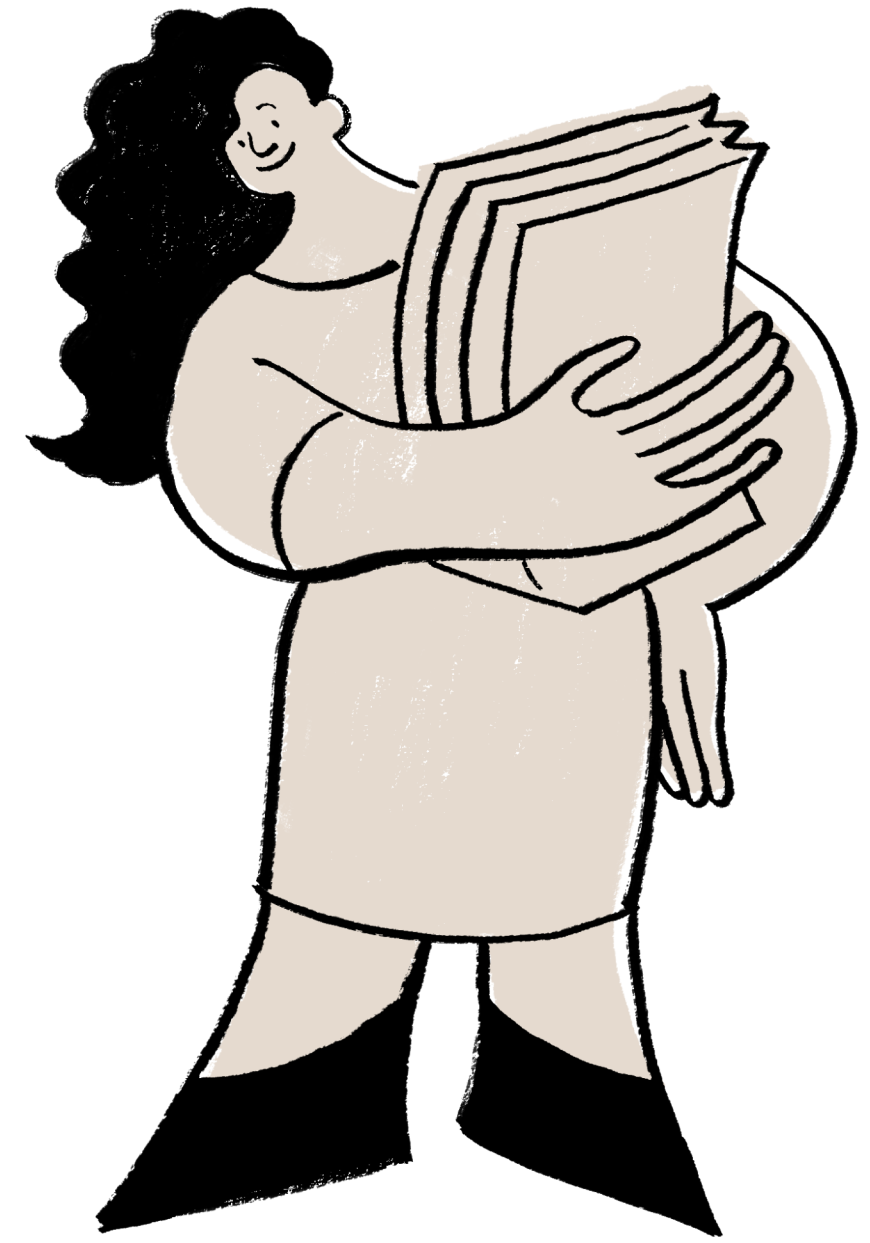
General Risk Strategy

- **Safe:** Low risk based on employment, age, income, and credit score.
- **Review Needed:** Moderate risk, e.g., young employed applicants with moderate income.
- **High Risk:** Unemployed, low external score, multiple previous rejections, high EMI.



THE TEAM

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THANK YOU

